SAFETY DATA SHEET

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

- **PRODUCT NAME : ACRYLIC ACID**

Name of the supplier	:	PT. NIPPON SHOKUBAI INDONESIA
Address	:	Jl. Raya Anyer km. 122, Ciwandan Cilegon 42447 Banten Tel No : 62-254-600660 (Hunting) Fax. No. : 62-254-600657 E-mail Address : sds@shokubai.co.id
Charge section	:	Safety Environment Department
Emergency phone number	:	62-254-600660 (Hunting)
Recommended use	:	Material of acrylic esters, Super absorbent polymers, Fiber modifier, Non-woven fabric binder
Restrictions on use	:	No information

2. HAZARD IDENTIFICATION

GHS CLASSIFICATION OF PRODUCT	***CATEGORY***
Flammable liquids	: Category 3
Pyrophoric liquids	: Not classified
Acute toxicity (oral)	: Category 4
Acute toxicity (dermal)	: Category 3
Acute toxicity (inhalation: vapour)	: Category 3
Acute toxicity (inhalation: dust /mist)	: Category 4
Skin corrosion/irritation	: Category 1A
Serious eye damage /eye irritation	: Category 1
Skin sensitization	: Not classified
Specific target organ systemic toxicity following single exposure	: Category 1
Specific target organ systemic toxicity following single exposure	: Category 2
Specific target organ systemic toxicity following repeated exposure	: Category 1
Acute hazards to the aquatic environment	: Category 1
Chronic hazards to the aquatic environment	: Category 2

GHS LABEL ELEMENTS

Pictograms :

Signal Word : Danger

HAZARD STATEMENTS

- Flammable liquid and vapour.
- Harmful if swallowed.
- Toxic in contact with skin.
- Toxic if inhaled.
- Harmful if inhaled
- Causes severe skin burns and eye damage.
- Causes serious eye damage.
- Causes damage to organs (respiratory apparatus, kidney).
- May cause damage to organs (liver).
- Cause damage to organs (respiratory apparatus) through prolonged or repeated exposure.
- Very toxic to aquatic life.
- Toxic to aquatic life with long lasting effects

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PRECAUTIONARY STATEMENTS

[Prevention]

- Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
- Keep container tightly closed.
- Ground and bond container and receiving equipment.
- Use explosion-proof [electrical/v entilating/l ighting] equipment.
- Use non-sparking tools.
- Take action to prevent static discharges.
- Do not breathe dust/f ume/g as/m ist/v apours/s pray.
- Wash hands and face thoroughly after handling.
- Do not eat, drink or smoke when using this product.
- Use only outdoors or in a well-ventilated area.
- Avoid release to the environment.
- Wear protective gloves/protective clothing/eye protection/face protection.

[Response]

- Immediately call a POISON CENTER/ doctor.
- Get medical advice/ attention if you feel unwell.
- Rinse mouth.
- Take off immediately all contaminated clothing and wash it before reuse.
- Collect spillage.
- IF SWALLOWED: Call a POISON CENTRE/ doctor if you feel unwell.

- IF ON SKIN: Wash with plenty of water.
- IF INHALED: Remove person to fresh air and keep comfortable for breathing.
- IF exposed or concerned: Call a POISON CENTER/ doctor.
- In case of fire : Use appropriate extinguishing media to extinguish.
- IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
- IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower].
- IF IN EYES: Rinse cautiously with water for several minutes.
- Remove contact lenses, if present and easy to do. Continue rinsing.

[Storage]

- Store Locked up.
- Store in a well ventilated place. Keep container tightly close.
- Store in a well ventilated place. Keep cool.

[Disposal]

- Dispose of contents/container in accordance with local/regional/national/international regulation.

Other hazards:

The product can be charged with static electricity.
Even though polymerization inhibitor is added, polymerization can be initiated by heating, direct sunlight, peroxides, and iron rust. Progress of polymerization can cause temperature increase and accelerated vapor pressure rise, resulting in explosion.

3. COMPOSITION / INFORMATION ON INGREDIENTS

Substance/Mixture : Substance

Chemical name	Content wt.%	CAS Number
Acrylic acid (Inhibited)	99.7	79-10-7

4. FIRST-AID MEASURES

- Inhalation :
  Remove person to fresh air and keep comfortable for breathing.
  Get medical advice/attention.
- Skin Contact :
  Immediately wipe away the attached material with cloths and the like rapidly.
  Wash with plenty of water.
  IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower].
  Immediately call a POISON CENTER/doctor.
- Eye Contact :
  Wash the eyes with clean water for at least several minutes (desirably 15 minutes or longer), (Remove the contact lenses if possible.) and get medical attention by an eye specialist immediately. During washing, fully open the eyelid with fingers and wash in such a way as to let water reach to every corner of the eyeball and eyelid.

- Ingestion :
  Do not induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. Get medical attention immediately.

5. FIRE-FIGTHING MEASURES

- Suitable Extinguishing Media :
  Small Fire: Chemical powder extinguisher, alcohol resistant foam fire extinguishing media, carbon dioxide, or water spray.
  Large Fire : Alcohol resistant foam fire extinguishing media, water spray or fog
- Unsuitable Extinguishing Media :
  Direct straight streams (may spread a fire)
- Specific Hazards of fire-fighting :
  Will be easily ignited by heat, sparks or flames.
  Containers may explode when heated. Fire will produce irritating, corrosive, and /or toxic gases. May polymerize explosively when heated or involved in a fire. Vapor explosion hazard indoors, outdoors or in sewers.
- Specific extinguishing methods :
  Keep upwind of fire.
  Use of water spray at a large fire, when another fire extinguishing media may be inefficient.
  Fight fire from maximum distance or use unmanned hose holders or monitor nozzles. Cool containers with flooding quantities of water until well after fire is out. For massive fire, use unmanned hose holders or monitor nozzles; if this is impossible, withdraw from area and let fire burn.
  Large Fire: Do not get water inside containers.
- Special protective actions for fire-fighters :
  In case of fire fighting, wear positive pressure self-contained breathing apparatus (SCBA) and chemical protective clothing. Withdraw immediately in case of rising sound from venting safety devices or discoloration of tank. Structural firefighter's protective clothing provides limited protection in fire situation ONLY; it is not effective in spill situations where direct contact with the substance is possible.

6. ACCIDENTAL RELEASE MEASURES

- Personal Precautions, protective equipment and emergency procedures :
  Promptly remove possible ignition sources from the vicinity.
  Use personal protection recommended in Section 8. Isolate the hazard area and deny entry to unnecessary and unprotected personnel.
- Environmental Precautions :
  Do not flush to sewer or allow entering waterways.
  This product has a strong bad smell and irritation. Therefore, take proper measures like informing the people in the neighboring area of the leakage occurrence.
  Surround this product with sandbags and earth/sand and cover it with an antistatic sheet to prevent diffusion (of smell).
- Methods and materials for containment and cleaning up :
  In the case of small amount, neutralize it with 5 - 10% sodium hydroxide solution, and wash with water. The waste water should be properly treated (incineration, activated sludge process). In the case of large amount, surround it with earth/sand, sandbags and others, cover it with an oil-resistant antistatic sheet to prevent generation of vapor, and recover it into proper containers like drums. Then, treat it as in the case of small amount.

7. HANDLING AND STORAGE

- Handling :
  - Technical measures:
    This product can be charged with static electricity. Take countermeasures for static electricity removal (grounding, others). Wear antistatic clothes and antistatic shoes to prevent human body electrification.
    Use explosion-proof [electrical/ventilating/lighting] equipment.
    Use only non-sparking tools.
    Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
    Do not handle the containers harshly in such a way as to make them tumbled, dropped, given a shock, or dragged.
  - Local/total ventilation:
    Use the ventilation equipment describe in Section 8.
  - Precautions for safe handling:
    Not especially.
  - Contact avoidance:
    Refer to the information in section 10 (Stability and reactivity).
  - Hygiene measure:
    During handling, be sure to wear proper protective equipment (refer to the section 8)
- Storage
  - Condition for safe storage:
    Keep away from heat.
    Keep away from sources of ignition – No smoking.
    Protect from sunlight. Store in a well-ventilated place.
    When storing in a tank, store this product at a temperature of 15-25^{0}C and control to keep the oxygen concentration of the gaseous phase at 5-21% to prevent polymerization. (Storage at on oxygen concentration of lower than 5% can cause polymerization.) Store the containers at 15-25^{0}C to prevent freezing and polymerization. When it is freezed, gradually and completely dissolve it with warm water of 40^{0}C or lower, while avoiding local heating because the polymerization inhibitor is distributed unevenly. Thoroughly agitate and homogenize the completely dissolved product to prevent uneven distribution of the polymerization inhibitor in the container.
  - Safe packaging material:
    Store in tightly closed container.
    Stainless steel or polyethylene container.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

- Permissible concentration

Chemical Name	ACGIH		U.S. - OSHA	
	TLV-TWA	TLV-STEL	PEL-TWA	PEL-STEL
Acrylic acid (Inhibited)	2 ppm Skin	Not established	Not established	Not established

- Appropriate engineering controls:
  Install eye and body washing facilities near the handling place. Use closed equipment/machinery or use local ventilation equipment.

- Personal protective equipment :

  Respiratory Protection:

  Use appropriate respiratory protection.
  Gas mask
  Consider wearing an air-supplied respirator when handling highly concentrated chemicals.
  When using a gas mask in an environment where workers are exposed to dust, use a combined filter for both hazardous gas and particle.
  Select a gas mask with performance and structure suitable for the work that complies with the laws and standards of the area where it is handled, referring to the data provided in the instruction manual.

  Hand protection:

  Use appropriate protective gloves.
  Organic solvent impermeable protective gloves (antistatic ones are desirable).
  For the selection of protective gloves, set a sufficient usage time for work by referring to the class of permeability described in the instruction manual, and use gloves within that time.

  Eye/face protection :

  Use appropriate protective eyewear.
  Protective glasses, goggle, protective face shield

  Skin and body protection :

  Use appropriate protective clothing and shoes.
  Organic solvent impermeable protective clothes, protective shoes (antistatic ones are desirable).

9. PHYSICAL AND CHEMICAL PROPERTIES

[Chemical product]

- Physical state : Liquid
- Form : Liquid
- Colour : Colorless
- Odour : Strong irritating odor like that of acetic acid
- Odour threshold : 0.1 (ppm)
- Melting point/freezing point : 13.5 (°C)
- Boiling point or initial boiling point and boiling range : 141 °C
- Flammability : No data
- Lower and upper explosion limit/ flammability limit : 2– 20.2 (vol%)
- Flash point : 51.4 (°C) Tag closed cup
  68 (°C) Open Cup
- Auto-ignition temperature : 428 °C
- Decomposition temperature : No data
- pH : 2.3 (20 °C 1mol/L aqueous solution)
- Viscosity : 1.25 (mPa▪S) (25 °C)
- Kinematic Viscosity : No data
- Solubility : water : very good
- Partition coefficient n-octanol/water (log value) : 0.46
- Vapour pressure : 413 (Pa) (20 °C)

- Density and/or relative density : 1.05 (20 °C)
- Relative vapour density : 2.5
- Particle characteristic : No data
- Other data : Conductivity:8.8E-08 S/m
  Heat accumulation storage test(UN recommendations):55℃<SAPT<=75℃
  (SAPT:Self-accelerating polymerization temperature)

10. STABILITY AND REACTIVITY :

- Reactivity Chemical stability :
  Refer to the information in "Possibility of hazardous reactions".
  Even though a polymerization inhibitor is added, heating, direct sunlight, peroxide, and iron rust can cause polymerization.
- Possibility of hazardous reactions :
  When storing at low-oxygen condition of oxygen concentration of lower than 5% at the gaseous phase, polymerization can occur. During the storage of acrylic acid, acrylic acid dimers (acryloyloxypropionic acid) are generated. The rate of generation of acrylic acid dimers depends on the temperature and the amount of water; the higher the temperature, the higher the rate of generation. When especially the temperature of acrylic acid is high, the reaction rate of acrylic acid dimers generation is large. The temperature rises by the heat of reaction, so that there is danger of the polymerization reaction and the explosion to depend on holding time. Furthermore, the rate of generation of acrylic acid dimers is higher in acrylic acid containing more water compared to that of purified acrylic acid.

- Conditions to avoid :
  Heating, contact with incompatible substances, ignition sources.
- Incompatible materials:
  Strong oxidizing substance, peroxides.
- Hazardous decomposition product :
  No information.

11. TOXICOLOGICAL INFORMATION

[Chemical product]

Acute toxicity, oral : Category 4
LD50: 33.5-3200 mg/kg [rat]
LD50: 193 mg/kg [rat]

Acute toxicity, dermal : Category 3
LD50: 300-600 mg/kg [rat]
LD50: 280-950 mg/kg [rabbit]
LD50:> 2000 mg/kg [rabbit]

Acute toxicity, inhalation : Not classified (Gas)
Category 3 (Vapour)
LC50: 3.6-14.4 mg/L/4h [rat]
LC50: 11.1 mg/L/1h [rat]
Category 4 (Dust/Mist)
LC50: 2.75-3.75 mg/L/4h [rat]

Skin corrosion/irritation : Category 1A

Serious eye damage/irritation	: Category 1
Sensitization, respiratory	: Classification not possible No test data on the product
Sensitization, skin	: Not classified
Germ cell mutagenicity	: Classification not possible No data available
Carcinogenicity	: Classification not possible IARC: 3 ACGIH: A4
Reproductive toxicity	: Classification not possible No data available
Specific target organ toxicity (Single exposure)	: Category 1 (respiratory apparatus, kidney) Category 2 (liver)
Specific target organ toxicity (Repeated exposure)	: Category 1 (respiratory apparatus)
Aspiration hazard	: Classification not possible No test data on the product

12. ECOLOGICAL INFORMATION

[Chemical product]

Ecotoxicity	
Acute hazard	: Category 1
Fish	: LC50 (96hr): 27 mg/L [Oncorhynchus mykiss] LC50 (96h): 222 mg/L [Brachydanio rerio]
Crustacea	: EC50 (48h): 95 mg/L [Daphnia magna]
Algae or other aquatic plants	: ErC50 (72hr): 0.13 mg/L [Scenedesmus] EC50 (72h): 0.04 mg/L [Desmodesmus] EC50 (96h): 0.17 mg/L [Pseudokirchneriella subcapitata]
Chronic hazard	: Category 2
Fish	: NOEC(45day): >= 10 mg/L [Oryzias latipes]
Crustacea	: No data
Algae or other aquatic plants	: NOEC (72hr): 0.016 mg/L [Scenedesmus]
Persistence and degradability	: Rapidly biodegradable
Bioaccumulative potential	: Low bioaccumulation
Mobility in soil	: No data
Hazardous to the ozone layer	: Classification not possible

13. DISPOSAL CONSIDERATION

Disposal Methods:
When waste material and waste water are to be treated, incinerate them or collect them into specified containers and entrust the disposal to a licensed industrial waste disposal company.
Do not use the used containers for other purposes like filling other substances. After treating the content according to the above description, the empty containers should be reused, recycled, or disposed through suitably qualified or licensed contractor, in accordance with national/local regulations.
The waste water can be treated also by an activated sludge method.

14. TRANSPORT INFORMATION

- ICAO/IATA
  - UN classification : Class 8 (Subsidiary hazard 3)
  - UN number : 2218
  - Proper shipping name : ACRYLIC ACID, STABILIZED
  - Packing group : II
- IMDG
  - UN classification : Class 8 (Subsidiary hazard 3)
  - UN number : 2218
  - Proper shipping name : ACRYLIC ACID, STABILIZED
  - Packing group : II
- Marine pollutant : Yes
- Transport regulations : Transport this product in accordance with local/regional/national/international regulations.
- Special precautions : At the time of discharge from a tank truck, be sure to ground to prevent static electricity charge and to completely empty the remaining liquid in the pipes.
  Load the containers in such a way as not to fall down, tumble, or being damaged. Cover the loaded cargo to prevent direct sunlight.
  (Reference information) Heat accumulation storage test(UN recommendations) 55℃< SAPT<= 75℃ (SAPT：Self-accelerating polymerization temperature)

15. REGULATORY INFORMATION

- Inventories

Japan (ENCS)	: Listed
Korea (KECL)	: Listed
Australia (AIIC)	: Listed
Canada (DSL)	: Listed
China (IECSC)	: Listed
EU (EINECS)	: Listed
New Zealand (NZIoC)	: Listed
USA (TSCA)	: Listed
Philippine (PICCS)	: Listed
Taiwan (TCSI)	: Listed

- Restriction at export (Indonesia) : Not Applicable

16. OTHER INFORMATION

- Date prepare : 01/Sept/98
  Date revised : 15/Jan/24
  Date review : 15/Jan/29
  Revise No : 10

  Disclaimer:
  - The described information is based on the latest material, information, and other data available at the time of preparation; however, it does not intend to guarantee anything like contents, physical/chemical properties, and danger/hazards. Also, the precautions mentioned above are for ordinary handling; therefore, for special handling, it is the obligation of each user of the product to take proper measures suitable for each handling condition. Take enough care for such matters.
  - This SDS is based on JIS Z7252: 2019 and JIS Z7253: 2019. If "Not classified" is provided in the GHS classification in Section2, it means that sufficient information is available to classify the product and sufficient evidence is found to determine that it is not applicable to any hazard categories defined by JIS Z7252. And another case is that it based on comprehensive judgment by experts.
  - Regulatory information with regard to this substance in your country or region should be examined by your own responsibility.